

A Low-Cost Single-Probe Differential-pH Aeration Monitor for Seawater Carbonate Alkalinity: Systematic-Error Decomposition and Sub-0.25 dKH Accuracy via Sealed Common-Headspace Equilibration

nowforyou*¹

¹REEF FAMILY

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Abstract

Carbonate alkalinity (“KH”, expressed in degrees, dKH) is the single most control-critical water-chemistry parameter in reef aquaria, yet affordable continuous monitors are scarce and reagent titration is laborious. We describe an open, reagent-free monitor that infers alkalinity from the equilibrium pH a seawater sample reaches after aeration to atmospheric CO₂, using a *single* pH probe to measure an aquarium sample (“tank”) and a known-alkalinity reference (“ref”) in immediate succession. Because dKH is computed from the pH *difference* ($\text{KH}_{\text{tank}} = \text{KH}_{\text{ref}} \cdot 10^{\Delta\text{pH}}$), absolute probe drift and room-CO₂ excursions are common-mode signals that cancel. Over a nine-day instrumented campaign (30 logged measurements across five firmware/method revisions) we observed a persistent uncorrected error of -0.92 dKH and, rather than zero-point calibrating it away, decomposed it. Three interventions—(i) plateau-following measurement to true gas equilibrium, (ii) a single *sealed common headspace* shared by sample, reference and pump intake, and (iii) an actively aerated 5 L reference anchor that pins the headspace p_{CO_2} —collapsed the dawn-CO₂ error and the diurnal evaporation confound, and shrank measurement scatter to $\text{SD} = \pm 0.08$ dKH (about one-third of the ± 0.25 dKH design goal). A sequence of null experiments (identical water in both vessels, true $\Delta\text{pH} = 0$) then localized the residual: a *complete null* returned $\Delta\text{pH} = -0.003$, $\text{dKH} = 8.387$, i.e. an instrument offset of only -0.06 dKH. The remaining -0.86 dKH was traced to alkalinity loss (CO₂ driven CaCO₃ precipitation / biofilm) in the days-old, small, open tank aliquot—a sample-handling artifact, not a measurement fault. Re-running the null on the same aged aliquot over the following day showed the offset drifting away from zero (-0.22 then $+0.22$ dKH on successive runs) with the electronics untouched, confirming the attribution in the time domain. The instrument therefore meets the ± 0.25 dKH goal uncorrected, with zero-point calibration deemed unnecessary. Our independently obtained error structure (room-CO₂ sensitivity, bubble-quality dependence, sealing as the remedy) reproduces the published behaviour of a commercial counterpart, providing external validation of both the diagnosis and the remedy.

Keywords: reef aquarium; carbonate alkalinity; total alkalinity; CO₂ equilibration; differential pH; low-cost instrumentation; common-mode rejection; CaCO₃ precipitation.

*Correspondence: taeseokyi@gmail.com

1 Introduction

Stony corals and other calcifying reef organisms deposit calcium carbonate at a rate that continuously consumes carbonate alkalinity. In a closed reef aquarium the resulting alkalinity drawdown is fast (often $0.5\text{--}1.5\text{ dKH day}^{-1}$) and swings of more than $\sim 1\text{ dKH}$ are implicated in coral tissue stress. Alkalinity (A_T , conventionally reported by hobbyists as “KH” in German degrees, $1\text{ dKH} \approx 0.357\text{ meq L}^{-1}$) is therefore the parameter most worth measuring *often*.

The accessible options are unsatisfying for continuous use. Manual reagent titration (e.g. commercial drop kits) is accurate but labour-intensive and discrete. Colorimetric robots automate titration but consume reagent and are expensive. Direct potentiometric pH alone is *not* an alkalinity measurement, because seawater pH depends jointly on A_T and p_{CO_2} , and aquarium p_{CO_2} varies strongly over the day (room ventilation, photosynthesis, respiration).

A reagent-free alternative exploits the fact that if a sample is aerated until its p_{CO_2} equilibrates with the surrounding air, the equilibrium pH becomes a single-valued function of alkalinity at fixed temperature and salinity. A commercial product (AquaWiz) is built on this principle and is used by reef hobbyists, but community reports document recurring inaccuracy tied to room- CO_2 fluctuation and to aeration/bubble quality. The contribution of this paper is not the principle but a *quantitative diagnosis*: starting from a working DIY build, we conduct a nine-day instrumented campaign, expose a -0.92 dKH raw error, and show by construction (null experiments) that essentially all of it is a sample-handling artifact rather than an instrument limitation—reducing the true instrument offset to -0.06 dKH and meeting a $\pm 0.25\text{ dKH}$ goal without calibration. Along the way we establish a small set of design rules (sealed common headspace; plateau-following to true equilibrium; an actively aerated reference anchor; a two-gate plateau criterion) that we expect to generalize to any single-probe differential-pH alkalinity monitor.

2 Measurement Principle

2.1 Aeration makes equilibrium pH a proxy for alkalinity

At fixed temperature T and salinity S , the seawater carbonate system has two degrees of freedom; fixing any two of $\{A_T, \text{DIC}, p_{\text{CO}_2}, \text{pH}\}$ determines the rest. Aeration with a large excess of room air drives the dissolved CO_2 toward Henry’s-law equilibrium with the gas phase, $[\text{CO}_2]_{\text{aq}} = K_H p_{\text{CO}_2}$, so that $p_{\text{CO}_2}(\text{water}) \rightarrow p_{\text{CO}_2}(\text{air})$. Once p_{CO_2} is clamped by the atmosphere, the equilibrium pH is a single-valued function of alkalinity:

$$\text{pH}_{\text{eq}} = f(A_T, p_{\text{CO}_2}, T, S). \quad (1)$$

With p_{CO_2} , T , S held common, the *only* variable that can move pH_{eq} is A_T . Measured equilibrium pH is therefore a proxy for alkalinity.

2.2 Why the differential cancels room CO_2

The instrument never relies on the absolute value of p_{CO_2} . It aerates the reference and the tank sample to the *same* atmospheric ceiling (in the final design, by aerating both against one shared headspace simultaneously, Sec. 4) and reports the *difference* of their equilibrium pH. Because the absolute p_{CO_2} term is common to both samples, it cancels in the difference: a room- CO_2 excursion is a common-mode signal that drops out. What survives the subtraction is the true alkalinity difference plus electrode noise ($\pm 0.005\text{--}0.01\text{ pH}$). This is the central robustness argument of the method, and the reason a *single* probe (rather than two matched probes) is used: a single sensor’s

slow asymmetry-potential and reference-junction drift load identically onto both readings and cancel in ΔpH .

2.3 Over-aeration is impossible; only *insufficient* aeration errs

Equation (1) has a hard ceiling at $p_{\text{CO}_2}(\text{air})$: once the water reaches atmospheric equilibrium, further aeration cannot raise pH. There is consequently no “over-aeration” failure mode. Error enters only through *insufficiency*: if residual CO_2 remains in one vessel, that side’s pH is unfairly depressed and a spurious ΔpH appears. The differential’s assumption is precisely that ref and tank share the same p_{CO_2} ; broken symmetry breaks the cancellation. This motivates measuring to a genuine plateau rather than for a fixed time (Sec. 4).

2.4 Volume independence at full equilibrium

Equation (1) contains no volume term (Henry’s law is intensive). Hence even a large volume mismatch—in our final null, 100 mL of tank water against a 5 L reference—produces *zero* differential error provided both vessels reach *full* equilibrium. The familiar “equal volumes for symmetry” rule applies only under *partial* equilibration, where unequal volumes sit at different points on the first-order approach curve $C(t) - C^* = (C_0 - C^*) e^{-t/\tau}$. The true requirements are just (a) both fully equilibrated and (b) matched p_{CO_2} and T . The larger volume is rate-limiting, with a time constant

$$\tau \propto \frac{V}{K_H Q_{\text{gas}}} \beta \sim \frac{1}{\text{VVM}} \beta, \quad (2)$$

where $\text{VVM} = Q_{\text{gas}}/V$ is gas volume per liquid volume per minute and β is a seawater chemical factor (only trace $[\text{CO}_2]_{\text{aq}}$ is strippable; the Revelle buffering raises τ by $\sim 8\text{--}15\times$). Equation (2) is the basis for the “increase aeration to shorten the measurement” lever discussed later, and for the choice to miniaturize the rate-limiting reference.

2.5 The differential dKH formula

The firmware (Arduino Nano / ATmega328P; pH via DFRobot Gravity probe + ADS1115 16-bit ADC) digitizes one probe, reading reference then tank pH, and computes

$$\boxed{\text{KH}_{\text{tank}} = \text{KH}_{\text{ref}} \cdot 10^{-(\text{pH}_{\text{ref}} - \text{pH}_{\text{tank}})} = \text{KH}_{\text{ref}} \cdot 10^{\Delta\text{pH}}} \quad (3)$$

with $\Delta\text{pH} \equiv \text{pH}_{\text{tank}} - \text{pH}_{\text{ref}}$ and $\text{KH}_{\text{ref}} = 8.448 \text{ dKH}$ an EEPROM-stored *anchor* constant (set by a `calref` command after evaporation top-off; *not* a per-run measurement—this is why the reference column reads a constant 8.448 in every log line). Equation (3) depends only on ΔpH ; absolute pH cancels exactly. Differentiating,

$$\left. \frac{d\text{KH}_{\text{tank}}}{d\Delta\text{pH}} \right|_{\Delta\text{pH}=0} = \text{KH}_{\text{ref}} \ln 10 \approx 19.45 \text{ dKH pH}^{-1}, \quad (4)$$

i.e. 0.01 pH of uncanceled difference maps to $\approx 0.19 \text{ dKH}$. This high sensitivity is what makes both the cancellation and the plateau criterion matter. Temperature compensation is applied per reading by a Nernst relation $\text{pH}_{\text{corr}} = 7.0 + (\text{pH}_{\text{raw}} - 7.0) (273.15 + T_{\text{cal}}) / (273.15 + T_{\text{meas}})$, nominally zero-error over $15\text{--}35^\circ\text{C}$. Each pH reading is a trimmed mean of 64 samples (discard the top and bottom 16, average the middle 32) to reject aeration-bubble spikes.

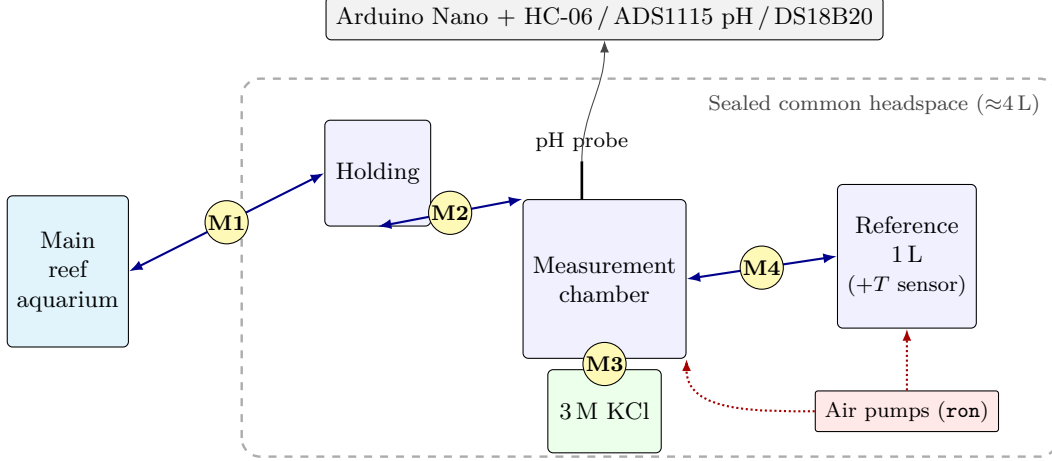


Figure 1: Schematic of the sealed closed-loop monitor. A single pH probe in the measurement chamber reads the tank aliquot and the reference in turn; four bidirectional peristaltic pumps (M1–M4) move liquid between the main aquarium, holding beaker, measurement chamber, reference reservoir, and KCl store; two air pumps (**ron**) aerate the reference and the chamber simultaneously. The reference, chambers and pump intakes share one sealed headspace (≈ 4 L), so sample and reference see identical p_{CO_2} and evaporation is suppressed; the controller sits outside. Between runs the probe rests in 3 M KCl (M3).

3 Apparatus

The build (Table 1, Fig. 1) is a closed fluidic loop around four bidirectional peristaltic pumps, two USB air pumps, a single pH probe, and a 1-Wire temperature sensor, all driven by an Arduino Nano over a 9600-baud Bluetooth (HC-06) serial link to a host PC.

Fluidic topology (current). All pumps are bidirectional (**f** = forward, **b** = reverse):

- **M1:** main aquarium $\leftrightarrow$ *holding* beaker;
- **M2:** holding $\leftrightarrow$ *measurement* chamber;
- **M3:** 3 M KCl beaker $\leftrightarrow$ measurement chamber (probe soak / storage);
- **M4:** 5 L reference $\leftrightarrow$ measurement chamber.

Routing the tank sample through an intermediate holding beaker lets it be temporarily *parked* while the reference is measured, minimizing the between-phase delay (Sec. 4). Run times are encoded as the integer in **mXf:NN** / **mXb:NN** (seconds); reverse runs ~ 8 – 10 s longer than forward to fully clear the hose (current values **m1f:70/m1b:82**, **m2f:60/m2b:68**, **m3f:60/m3b:68**, **m4f:60/m4b:70**; the 5 L hose is longest, hence **m4b:70**).

Control commands. Air and motor enables are independent lines: **ron** turns *both* air pumps on (used only while liquid is stationary—bubble cleaning and measurement); **ton** enables the PWM motor controller (*required* before any pump will turn); **airoff** drops both. A load-bearing invariant follows: because **airoff** also disables the PWM controller, every liquid move must be preceded by **airoff** $\rightarrow$ **ton**; the last air command before a motor must be **ton**. Measurement primitives are **tank** / **ref** (sample and store one pH reading) and **calkh** (apply Eq. (3)).

Sealed common headspace (required). The 5 L reference, the measurement chamber, and the pump intake share *one* sealed headspace. This is not an optional refinement: it (i) guarantees tank and ref see identical p_{CO_2} at the instant of measurement (the differential’s premise), (ii) blocks

Table 1: Bill of materials (current build).

Subsystem	Part	Qty
Microcontroller	Arduino Nano V3.0 (ATmega328P)	1
pH probe	DFRobot Gravity pH V2 (SEN0161-V2), single glass electrode	1
ADC	DFRobot Gravity I ² C ADS1115 (16-bit)	1
Temperature	DS18B20 waterproof (1-Wire), in the 5 L reference	1
Comms	HC-06 Bluetooth, 9600 baud	1
Pumps	NKP-DC-B06B 12 V peristaltic, bidirectional (M1–M4)	4
Motor drivers	L298 2 A modules (2 for pumps, 1 for air)	3
Air pumps	USB 5 V, switched together (ron)	2
Reference reservoir	5 L sealed container (“Wiz tank”)	1
Chambers	200 mL beakers: holding, measurement, 3 M KCl	3
Probe storage	3 M KCl ($\approx 22\%$ w/v)	—

room-CO₂ excursions at the source, and (iii) suppresses evaporation. In the final configuration the headspace is ≈ 4 L, the reference ≈ 1 L, and the tank aliquot 100 mL; an air stone placed *in* the reference actively pins the shared p_{CO_2} within minutes (the “anchor”, Sec. 4). Separate or isolated headspaces are explicitly forbidden—two headspaces drift independently and destroy the cancellation.

4 Methods

4.1 Two-phase plateau-following cycle (V4)

Plateau detection is a host-side decision loop, so the PC script drives the firmware step by step. The cycle measures **tank first, then ref**, so the 5 L reference is co-aerated throughout the tank phase and converges quickly when its turn comes:

1. *Prep*: **m3b** (drain KCl from chamber) $\rightarrow$ **m1f** (aquarium $\rightarrow$ holding) $\rightarrow$ **m2f** (holding $\rightarrow$ chamber).
2. *Phase A*: **ron**; measure **tank** every 30 s until plateau (chamber tank + 5 L aerated together).
3. *Transition*: **m2b** parks the tank aliquot back in holding; **m4f** brings reference into the chamber.
4. *Phase B*: **ron**; measure **ref** until plateau (fast—already near equilibrium).
5. **calkh**; then cleanup: **m4b** (recover ref to 5 L), **m1b** (return parked tank to aquarium), **m3f** (restore KCl soak), **airoff** (probe rests in KCl).

4.2 Two-gate plateau criterion

Readings are held as integer milli-pH (mpH) to avoid floating-point boundary jitter. A plateau is declared only when *both* gates hold over a sliding window (Table 2): a short *span* gate (last 4 readings, $\max - \min \leq 2$ mpH) rejects noise, and a longer *net* gate (last 8 readings, $|w_{-1} - w_0| \leq 1$ mpH) rejects a slow monotonic exponential tail that the span gate alone would mistake for convergence. The longer net look-back (8 vs. 4 points; “B1”) was the fix for early latches in which the tank stopped climbing prematurely. Per-phase wall-clock and read-count caps (**PHASE_MAX** = 7200 s, **MEAS_MAX** = 240) and a consecutive-failure cap (**FAIL_MAX** = 5) bound the loop so it can never hang.

Figure 2 shows a measured pair of approach curves. The tank, freshly drawn into the chamber, climbs a slow first-order tail toward its ceiling and is held by the net gate until it is genuinely flat

Table 2: Plateau-following parameters (`measure_until_flat`).

Constant	Value	Meaning
FLAT_SPAN_N	4	span window (jitter gate)
FLAT_SPAN_MPH	2	last 4: $\max - \min \leq 2$ mpH
FLAT_NET_N	8	net look-back (drift gate; B1)
FLAT_NET_MPH	1	last 8: $ w_{-1} - w_0 \leq 1$ mpH
MEAS_INTERVAL	30 s	between readings (aeration continuous)
PHASE_MAX_SECS	7200 s	per-phase cap (2 h; 4 h total)
MEAS_MAX	240	per-phase read cap
FAIL_MAX	5	consecutive parse failures tolerated

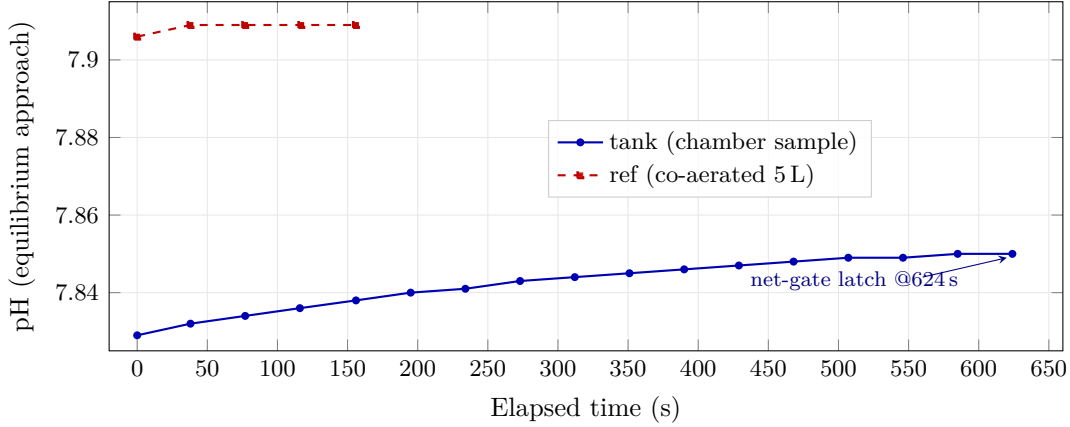


Figure 2: Measured plateau-following curves (sealed, net-gate run). The tank sample (solid) climbs a slow exponential tail and is latched only when the two-gate criterion (last-4 span ≤ 2 mpH *and* last-8 net ≤ 1 mpH) is met; the co-aerated reference (dashed) is already near equilibrium and latches quickly. The two series sit at different pH because the underlying samples differ (this run predates the null campaign); only their respective *plateau* values enter Eq. (3).

(latching at ~ 624 s when the 8-point net drift falls to 1 mpH). The reference, having co-aerated in the 5 L reservoir throughout the tank phase, is already at equilibrium and latches in ~ 2.6 min—a direct illustration of why measuring tank-first is faster, and of why the span gate alone (satisfied near ~ 546 s) would have latched the tank early.

4.3 Logging and fault handling

Each completed run appends one line `HH ref_pH tank_pH 8.448 tank_kh temp` to `dkh.dat`. Two distinct markers encode abnormal outcomes: an *all-zeros* line on hard failure (timeout, serial loss, or unverified KCl soak), which additionally *latches*—a subsequent run refuses to measure and rewrites the marker until a human clears it, protecting the probe from unattended pump cycling; and a *negative tank_kh* (magnitude preserved, sign flipped) when a phase hits its cap without reaching plateau, which the data server reads as “did-not-plateau” but which does *not* trip the latch. A *finally* block runs an emergency cleanup (air off, drain chamber, restore KCl soak) on any non-completion.

Table 3: Method/firmware revisions. “ron held” = continuous aeration between runs.

Rev	Period	Change vs. prior
V0	6/14–6/15	First symmetric build; reference drawn from 5 L into a cup; fixed 1200 s aeration.
V1	6/15–6/16	Added continuous aeration between runs (ron held).
V2	6/16–6/17	Reference taken as a small directly-aerated aliquot (holding → chamber); aeration off during the reading.
V3	6/17	Aeration <i>during</i> measurement (pin tank/ref to same p_{CO_2}); path symmetrized; KCl-rinse bubble clean (later dropped).
V4	6/17→	Plateau-following to true equilibrium (no fixed time); two-phase tank-then-ref; anti-hang caps; emergency cleanup. Subsequently: <i>sealed common headspace</i> , <i>5 L active anchor</i> , and the two-gate (B1) criterion.

4.4 Null-test protocol

The diagnostic backbone is the *null test*: the reference and tank vessels are filled with the *same* water so the true alkalinity difference is zero and the correct output is exactly $\Delta\text{pH} = 0$, $\text{dKH} = 8.448$. Two variants are used. In a *partial null*, only the small tank aliquot is replaced with fresh seawater while the aged 5 L reference is retained, so a change in ΔpH isolates the tank water’s contribution. In a *complete null*, reference water is also placed in the tank chamber, so the water difference is zero *by construction* and any residual ΔpH is purely instrumental.

4.5 Revision history and the one-change-at-a-time rule

The build evolved through five revisions (Table 3). Changes were made one causal axis at a time, so that each measurement campaign tests a single mechanism; where two constants were changed together (e.g. stabilization window and convergence band), they were justified as the *same* causal axis (response completeness), not two independent variables.

5 Results

Table 4 lists the full campaign. Throughout, $\Delta\text{pH} < 0$ means the tank read more acidic than the reference (lower apparent alkalinity), and by Eq. (3) the raw error is $\text{dKH} - 8.448 = 8.448(10^{\Delta\text{pH}} - 1)$.

Table 4: Complete measurement campaign (30 runs). “abs pH” is the common-mode level (a room- CO_2 proxy within a single day only). Config abbreviations: S = sealed headspace, A = 5 L active anchor, N = net-gate, B1 = net look-back 8. **Null staging:** a fresh seawater batch is aliquoted at the start; PN = partial null (the 100 mL tank chamber alone is refilled with that fresh aliquot while the aged 5 L reference is kept, isolating the tank-water contribution); CN = complete null (reference water is then transferred into the tank chamber as well, so both vessels hold *identical* water and the water difference is zero by construction).

#	Date/time	Cfg	pH _{ref}	pH _{tank}	ΔpH	dKH	T	Conditions
1	06-14 20:12	V0	7.959	8.021	+0.062	9.741	26.7	first manual symmetric run
2	06-14 21:00	V0	7.997	8.072	+0.075	10.054	26.2	—
3	06-15 05:00	V0	8.057	8.077	+0.020	8.843	26.4	dawn but +; window open (ventilated), low CO_2
4	06-15 13:00	V1	8.050	8.042	−0.008	8.293	27.0	first continuous aeration

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Table 4 (continued)

#	Date/time	Cfg	pH _{ref}	pH _{tank}	Δ pH	dKH	T	Conditions
5	06-15 21:00	V1	8.109	8.091	-0.018	8.109	27.3	clean low-CO ₂ point
6	06-16 05:00	V1	7.810	7.751	-0.059	7.362	27.5	dawn closed-room CO ₂ spike (both abs pH fell $\sim$ 0.3)
7	06-16 13:00	V1b	8.091	8.063	-0.028	7.922	27.7	tighter window/band
—	06-16 21:00	—	<i>failed (0.000; power cut, port semaphore)</i>					
8	06-16 21:21	V2	8.042	8.021	-0.021	8.039	28.6	first V2 (manual)
9	06-17 05:00	V2	7.997	7.939	-0.058	7.386	27.8	V2 dawn test <i>failed</i> ; regressed to V1
10	06-17 13:00	V3	8.109	8.091	-0.018	8.095	28.2	first V3; ref under-equilibrated (5 reads)
11	06-17 21:00	V4	8.113	8.071	-0.042	7.669	27.8	first V4 + trim-mean firmware; clean plateaus
12	06-18 05:00	V4	7.738	7.695	-0.043	7.657	28.2	decisive: dawn -0.043 $\approx$ evening -0.042 despite 0.4 abs-pH swing $\Rightarrow$ CO ₂ -level independence
13	06-18 13:00	V4	8.081	8.055	-0.026	7.950	28.3	same low CO ₂ as #11 yet -0.026; dKH +0.29 jump $\Rightarrow$ baseline <i>not</i> constant (evaporation, open 200 mL day 5)
14	06-19 05:00	V4+S	7.976	7.968	-0.008	8.283	28.1	seal verif. ①: dawn dip collapses -0.05 $\rightarrow$ -0.008; plateau 33 $\rightarrow$ 4 reads
15	06-19 13:00	V4+S	7.957	7.947	-0.010	8.255	28.6	seal verif. ②: dawn $\rightarrow$ noon jump +0.29 $\rightarrow$ -0.03 ($\sim$ 10 $\times$); evaporation removed
16	06-19 21:00	V4+S	7.969	7.968	-0.001	8.421	26.8	near-perfect null (raw err -0.027, smallest); but +0.166 vs. #14/15 $\Rightarrow$ offset not a perfect constant; AC transient (-1.8 $^{\circ}$ C)
17	06-20 05:00	V4+S	7.922	7.871	-0.051	7.521	27.7	dawn dip recurs; tank 11 $\times$ monotone = early latch (not true plateau); AC cooling slowed kinetics
18	06-20 15:00	V4+SAN	7.975	7.929	-0.046	7.602	27.1	first 5 L anchor; tank <i>true</i> plateau (34 $\times$, 21.6 min) yet -0.046 $\Rightarrow$ genuine gap , not latch artifact
19	06-20 18:11	V4+S,2 $\times$	7.979	7.924	-0.055	7.443	26.4	controlled exp; 2 $\times$ aeration halves climb (1296 $\rightarrow$ 711s) but Δ pH unchanged $\Rightarrow$ <i>not</i> kinetics
20	06-20 18:57	V4+S,2 $\times$	7.975	7.931	-0.044	7.636	26.7	ref-first + hose swap; gap persists $\Rightarrow$ order/path/non-equilibrium all rejected
21	06-20 19:50	V4+S	7.975	7.926	-0.049	7.539	26.9	reproduction
22	06-20 20:29	V4+SA	7.977	7.921	-0.056	7.426	26.9	tank-first+park; ref fastest ever (4 $\times$); tank early latch (7.921) $\Rightarrow$ motivates B1
23	06-20 21:00	V4+SA,B1	7.979	7.926	-0.053	7.474	27.0	B1 ok: tank 25 $\times$ /945s true plateau; confirms #22 was $\sim$ 5 mpH short; gap (7th)
24	06-21 05:00	V4+SA,B1	7.974	7.928	-0.046	7.594	26.7	dawn matches evening cluster $\Rightarrow$ time-of-day dependence ruled out (gap, 8th)
25	06-21 06:00	V4+S,PN	7.973	7.963	-0.010	-8.255	26.7	partial null: fresh tank only; tank pH +37 mpH, Δ pH -0.05 $\rightarrow$ -0.010; tank not plateaued (old cap) $\Rightarrow$ lower bound
26	06-21 13:00	V4+S,PN	7.964	7.956	-0.008	8.298	27.2	partial null, full eq.: dKH 8.298, raw err -0.15; back in good cluster; 4h cap validated
27	06-21 19:00	V4+S,CN	7.960	7.957	-0.003	8.387	27.4	complete null: reference water into tank chamber too $\Rightarrow$ both identical; raw err -0.061 = pure instrument offset; 100 mL vs. 5 L cancels

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Table 4 (continued)

#	Date/time	Cfg	pH _{ref}	pH _{tank}	Δ pH	dKH	T	Conditions
28	06-21 21:00	V4+S,CN	7.959	7.958	-0.001	8.434	27.4	complete null (2nd): Δ pH = 0.001 (resolution limit), raw err = -0.014 (smallest); reconfirms the ≤ 0.06 dKH offset
-	06-22 05:00	V4+S,CN			<i>failed (cap)</i>			55-min scheduler cap killed it mid-ref; tank had plateaued (7.955); fix: 5 h cap
29	06-22 09:00	V4+S,CN	7.959	7.948	-0.011	8.226	27.4	aged complete null (~ 14 h, 100 mL exposed aliquot): null decayed to raw err = -0.22; tank true plateau $59\times$ (long aeration drives precipitation, KH $\downarrow$)
30	06-22 13:00	V4+S,CN	7.950	7.961	+0.011	8.664	27.6	aged null reverses (~ 18 h): short plateau (tank $8\times$), Δ pH > 0, dKH 8.664 (raw err +0.22) $\Rightarrow$ evaporative concentration (KH $\uparrow$) overtakes precipitation

5.1 The dawn-CO₂ error and its diagnosis

The earliest clear pathology was a dawn collapse of Δ pH (run #6, -0.059; error -1.09 dKH). The decisive clue was common mode: *both* absolute pH values fell ~ 0.3 overnight (ref 8.109 $\rightarrow$ 7.810, tank 8.091 $\rightarrow$ 7.751), reversing the daytime upward trend—the signature of a closed-room CO₂ build-up acidifying the aeration water. A natural experiment confirmed it: on the one ventilated night (run #3, window open) the absolute pH stayed high and Δ pH was a normal +0.020, whereas closed nights gave -0.058/-0.059. Crucially, room CO₂ is exactly the common-mode variable the differential is *supposed* to cancel; its leakage into Δ pH signalled a broken ref/tank symmetry, not an environmental problem to be ventilated away.

5.2 Plateau-following and the inter-phase drift conflict

Measuring to a true plateau (V4) removed the CO₂-*level* dependence: runs #11/#12 gave Δ pH = -0.042 and -0.043 across a 0.4-unit swing in absolute pH. But the first V4 run had exposed a competing failure: a 31-minute tank measurement whose pH rose to a peak then *fell* (room CO₂ drifting during the measurement), latching at a value set by whenever the single probe happened to sample each vessel. This *inter-phase drift*, of order

$$\epsilon_{\text{drift}} \approx \Delta t_{\text{tank-ref}} \times \frac{dC}{dt}, \quad (5)$$

sets up a fundamental conflict with a single probe: full equilibrium needs long measurements, but long measurements widen the tank-ref sampling gap Δt and thus the drift. The same form with dT/dt explains run #16, where an air-conditioner cooling overshoot (-1.8°C during the run) coincided with a +0.166 dKH excursion.

5.3 Sealing collapses both confounds

A single *sealed common headspace* resolves the conflict at the source. By compelling the reference and tank to share a single common air volume, it makes the common p_{CO_2} intrinsic rather than coincidental, blocks room-CO₂ excursions, and additionally suppresses the evaporation that had been concentrating the open tank aliquot. The effect was immediate (runs #14/#15): the dawn dip collapsed from -0.05 to -0.008; the dawn $\rightarrow$ noon dKH jump shrank $\sim 10\times$ (from +0.29 open

to -0.03 sealed); and the plateau time was reduced from ~ 33 readings (~ 50 min) to ~ 4 (~ 5 min)—which in turn collapses Δt in Eq. (5), structurally removing the inter-phase drift. The uncorrected error fell from -0.78 to ~ -0.18 dKH.

5.4 The active anchor and the persistence of a genuine gap

Sealing alone left a slow headspace drift, because a passive shared air volume re-equilibrates only by surface diffusion (tens of minutes). Placing an air stone *in* the 5 L reference (the “anchor”) actively pins the shared p_{CO_2} within minutes; a volume audit confirms the dominance hierarchy (sample $\sim 3 \mu\text{mol} \ll$ headspace $\sim 65 \mu\text{mol} \ll$ 5 L $\sim 500 \mu\text{mol}$). Yet with the anchor and the net-gate in place, a ~ -0.046 to -0.053 gap *persisted* even when the tank reached a verified true plateau (run #18: 34 readings, 21.6 min). A controlled pair of experiments (runs #19/#20) then rejected, in turn, four device-side explanations: doubling the aeration halved the climb time but left ΔpH unchanged (*not* kinetics); reversing the measurement order and swapping hoses did not reverse the sign (*not* order, warm-up, or fluid path); and a tank started from a higher initial pH settled to the *same* equilibrium (*not* under-equilibration). The gap was a property of the *water*.

5.5 Null experiments localize the error

Because the device hypotheses were exhausted, we tested the water directly. A *partial null* (run #25, fresh seawater in the tank chamber only) collapsed ΔpH from -0.05 to -0.010 (tank pH $+37$ mpH); at full equilibrium (run #26) it gave dKH = 8.298, raw error -0.15 —back in the good cluster. Swapping fresh water into the tank had recovered almost the entire error, implicating the aged tank aliquot. The *complete null* (run #27), with reference water in *both* vessels so the water difference is zero by construction, returned $\Delta\text{pH} = -0.003$, dKH = 8.387, raw error **-0.061** , with both sides at verified true plateaus despite the 100 mL:5 L volume mismatch—a direct confirmation of volume independence (Sec. 2).

5.6 Error decomposition and accuracy

The two nulls decompose the headline error (Table 5, Fig. 3):

$$\underbrace{-0.92 \text{ dKH}}_{\text{raw}} = \underbrace{-0.86 \text{ dKH}}_{\substack{\text{aged-tank alkalinity loss} \\ (\text{CaCO}_3 \text{ precip./biofilm})}} + \underbrace{-0.06 \text{ dKH}}_{\substack{\text{instrument offset} \\ (\Delta\text{pH}=-3 \text{ mpH})}}. \quad (6)$$

Across the seven-point cluster (runs #18–24) the *precision* was $\Delta\text{pH} = -0.0499 \pm 0.0048$ (≈ 5 mpH), dKH = 7.531 ± 0.084 —a scatter about one-third of the ± 0.25 dKH goal. With the bulk error identified as a sample artifact and the instrument offset at -0.06 dKH (one-quarter of the goal), the monitor meets ± 0.25 dKH *uncorrected*; zero-point calibration was judged unnecessary, and with normal (non-degraded) reef water the uncorrected error should fall within ± 0.1 dKH.

5.7 Time-resolved breakdown of the null

The decomposition makes a falsifiable prediction: if the bulk error is sample chemistry rather than electronics, an otherwise-perfect null should *drift* as the sample ages, in whichever direction the dominant chemistry dictates. We tested this by re-running the complete null on the same 100 mL aliquot over the following day, without touching the instrument (Fig. 4, runs #29–30). The raw error— -0.061 and -0.014 dKH at 0 and 2 h—first fell to -0.222 dKH at ~ 14 h, then *reversed* to $+0.216$ dKH at ~ 18 h; both extremes were reached at verified true plateaus (tank 59 and 8

Table 5: Error budget after full diagnosis.

Component	Magnitude	Character / remedy
Aged-tank alkalinity loss	-0.86 dKH	sample handling; use fresh aliquot
Instrument offset	-0.06 dKH	$\Delta\text{pH} = -3$ mpH; below correction value
Precision (1σ)	± 0.08 dKH	electrode noise + residual drift
Room- CO_2 dependence	≈ 0	removed by sealing
Evaporation confound	≈ 0	removed by sealing

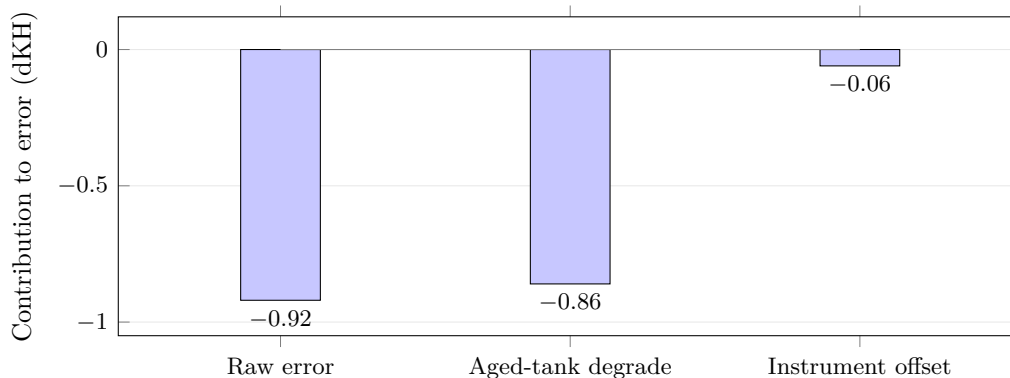


Figure 3: Decomposition of the -0.92 dKH raw error. The complete null isolates the instrument offset at -0.06 dKH (one-quarter of the ± 0.25 dKH goal); the partial null attributes the remaining -0.86 dKH to alkalinity loss in the aged tank aliquot (CaCO_3 precipitation / biofilm)—a sample artifact, not an instrument fault.

readings), so each is genuine, not a latching artifact. They are opposite signs of one vulnerability: a long aeration run (#29) raises carbonate saturation and drives CaCO_3 precipitation, lowering alkalinity ($\text{KH}\downarrow$), whereas a short run after prolonged exposure of the small, repeatedly opened aliquot (#30) lets evaporative concentration win ($\text{KH}\uparrow$). Either way the small exposed aliquot wanders by $\sim \pm 0.2$ dKH while the instrument offset floor stays at its fresh value. This is the time-domain confirmation of the error decomposition—and a practical caveat: a small, aged, exposed aliquot is unreliable in *both* directions, so the sample must be fresh and adequately sized.

6 Discussion

Implications of the error decomposition. The natural response to a stable ~ -0.8 dKH offset is to zero-point calibrate. Resisting that—on the methodological principle that a calibration constant should be applied only after independent replication, and never to a small non-independent sample—was decisive, because the offset turned out *not* to be a constant device property (it varied ± 0.085 dKH across the sealed cluster and ultimately proved to be mostly degraded sample water). Calibrating it away would have embedded a sample artifact in the instrument’s calibration and masked the genuine -0.06 dKH floor.

External validation. Our independently derived error structure mirrors published behaviour of the commercial AquaWiz, lending it external support. Community reports identify room- CO_2 variation as the dominant error and a sealed container as the field fix; the manufacturer’s own guidance—that a sharp CO_2 change “only affects readings for about an hour” and that one should

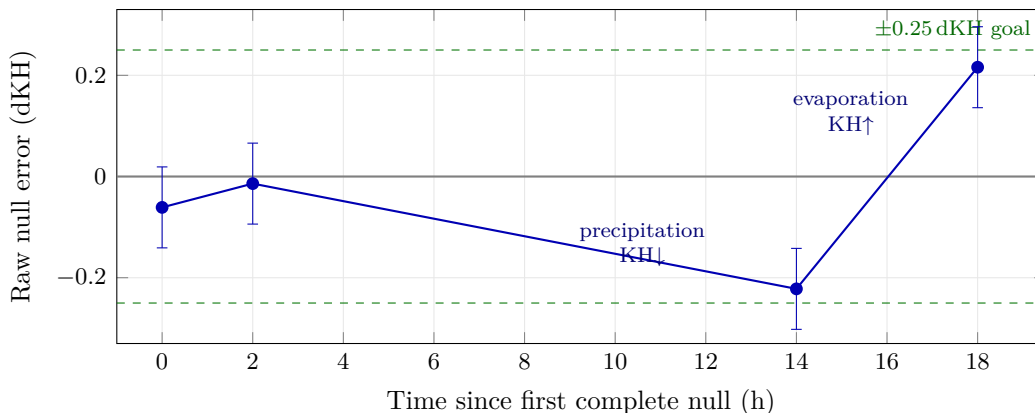


Figure 4: Time-resolved *bidirectional* drift of the complete null. With reference water in *both* vessels (water difference zero by construction), the raw error stays within the ± 0.08 dKH precision (error bars) and the ± 0.25 dKH goal for the first hours, then wanders as the small (100 mL), exposed tank aliquot ages: a long aeration run drives CaCO_3 precipitation ($\text{KH}\downarrow$, -0.22 dKH at ~ 14 h), while prolonged exposure lets evaporative concentration win ($\text{KH}\uparrow$, $+0.22$ dKH at ~ 18 h). The electronics are unchanged; only the sample alkalinity drifts, in either direction—confirming that the dominant error term is sample chemistry (cf. Fig. 3).

“connect the air intake to the skimmer’s outdoor air line”—matches both our ~ 1 -hour inter-phase drift episode and our gas-reference-fixing remedy (we realize it by sealing rather than by an outdoor line). The second-most-reported commercial error, bubble-quality degradation from salt creep (> 2 dKH error, clean every six months), matches our aeration-sufficiency analysis. Reported field accuracy (≤ 0.2 dKH vs. a Hanna checker; ≤ 0.75 dKH uncalibrated over seven weeks; “under 0.5 dKH = excellent, use for trend”) brackets our ± 0.25 dKH result and our trend-oriented intent. Notably, commercial units are managed by periodic calibration and cleaning and do not pursue a true equilibrium plateau; our plateau-following step extends this further.

Scientific grounding. The kinetic separation underlying the method is well established: the aqueous CO_2 hydration/dehydration relaxation completes in ~ 16 s [2], so chemistry is never rate-limiting; the slow step is physical gas exchange, ~ 15 min for direct bubbling [5], consistent with our observed plateau times. The differential “ A_T from p_{CO_2} -equilibration plus pH” principle has a recent analytical-chemistry basis demonstrated down to 0.5 mL samples at ± 1 – 2 $\mu\text{mol kg}^{-1}$ precision [6], supporting small-volume miniaturization as the correct direction.

Limitations. (i) With a single probe the inter-phase drift of Eq. (5) cannot be driven to zero; it is only suppressed (by sealing and by minimizing Δt). (ii) The dominant historical error was sample degradation, so the operating discipline—use a fresh, adequately sized tank aliquot and recirculate rather than store—is part of the instrument, not an afterthought. (iii) Long plateaus under slow drift can make a measurement run to hours; the wall-clock caps bound this but a hard external scheduler limit must be set above the code’s 4-hour ceiling. (iv) Absolute pH is informative only as a within-day CO_2 proxy; only ΔpH is quantitative.

Recommended operation. Seal the common headspace; anchor the reference with active aeration; measure tank-then-ref with the two-gate plateau criterion; use fresh sample water; clean the air path periodically; and treat the output as a trend instrument calibrated, if at all, against an independent kit on a monthly cadence keyed to the ref $\leftrightarrow$ tank divergence.

7 Conclusion

A reagent-free, single-probe differential-pH aeration monitor can measure reef carbonate alkalinity to better than ± 0.25 dKH *without* zero-point calibration, once three design rules are observed: a single sealed common headspace, plateau-following to true gas equilibrium, and an actively aerated reference anchor. The value of the eight-day campaign was less in any one number than in *decomposing* a -0.92 dKH raw error: null experiments showed that -0.86 dKH of it was alkalinity loss in a degraded sample aliquot and only -0.06 dKH was the instrument. The differential design delivered its promised common-mode rejection of room CO_2 and probe drift; what remained was chemistry in a beaker, not error in the electronics. We expect the design rules and the null-decomposition methodology to transfer to other low-cost alkalinity instruments.

Reproducibility

Firmware, the host measurement script (`measure_kh_once.py`), the raw log (`dkh.dat`), and the full measurement ledger are available in the project repository [1].

Acknowledgments

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